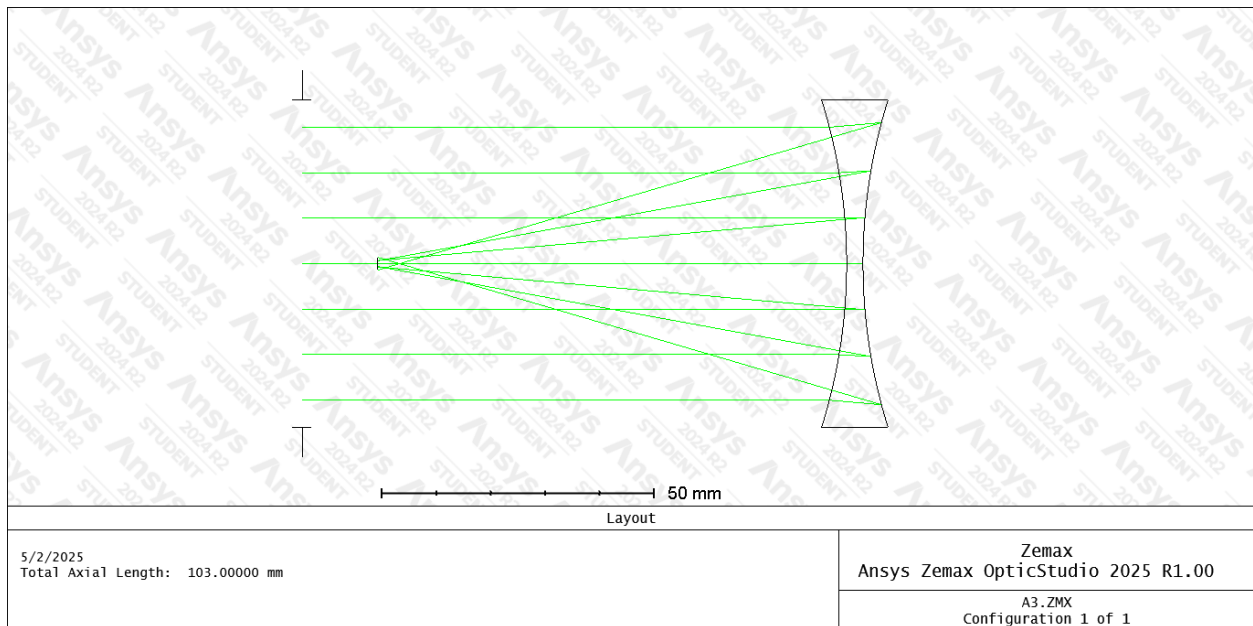


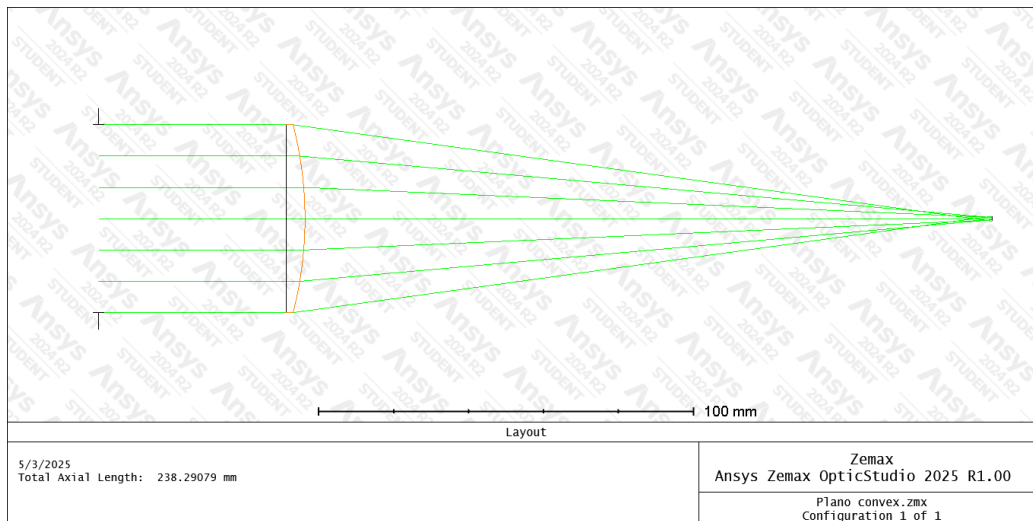
Assignment 1

- 1) Modify the **Bi-Convex Lens** by changing it to a **Bi-Concave Lens**.
- 2) Maintain the same **focal length (100 mm)** and **diameter (2 inches)**.
- 3) Perform a **ray trace analysis** to observe the beam divergence.
- 4) Save the results for further analysis.



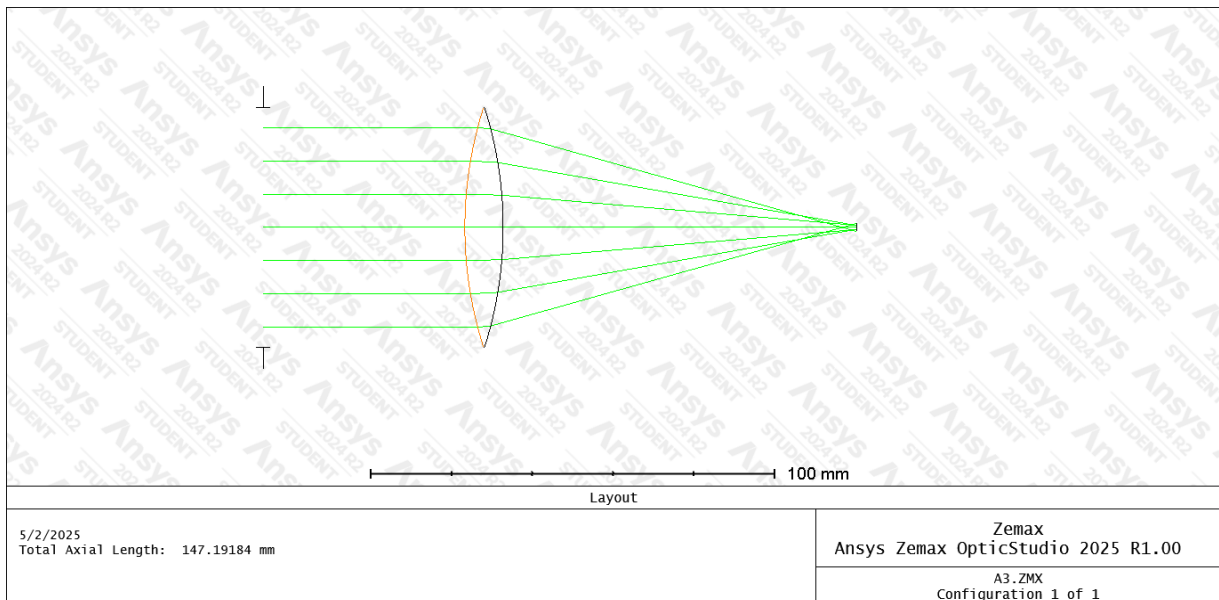
Assignment 2

1. Open **Zemax OpticStudio** and create a **new sequential optical system**.
2. Set up the **lens parameters**:
 - i. **Lens Type**: Plano-Convex
 - ii. **Focal Length**: 100 mm
 - iii. **Diameter**: 2 inches (~50.8 mm)
 - iv. **Material**: N-BK7
 - v. **Wavelength**: 550 nm (default)
3. Place a **point source** at the focal point of the lens to check collimation.
4. Perform a **ray trace analysis** and examine the beam profile.
5. Save the design for comparison in later assignments.



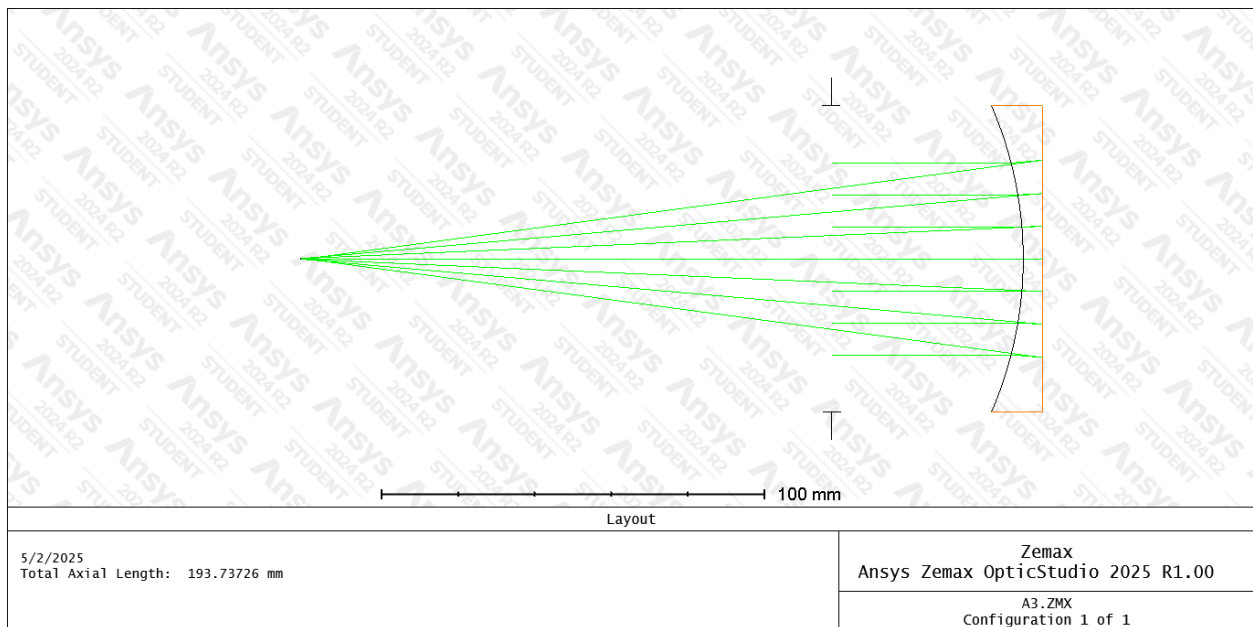
Assignment 3

1. Modify the existing system or create a new one.
2. Define the **lens parameters**:
 - i. **Lens Type**: Bi-Convex
 - ii. **Focal Length**: 100 mm
 - iii. **Diameter**: 2 inches
 - iv. **Material**: N-BK7
3. Perform **ray tracing** and compare the beam focusing behavior with the plano-convex lens.
4. Save the design.



Assignment 4

1. Modify the **Plano-Convex Lens** by changing its curvature to create a **Plano-Concave Lens**.
2. Keep the same **focal length (100 mm)** and **diameter (2 inches)**.
3. Perform **ray tracing** and observe the diverging nature of the beam.
4. Compare the results with the convex lenses.



Assignment 5

Design a simple afocal telescope system using Zemax (Sequential Mode). The telescope should meet the following core optical

Requirements:

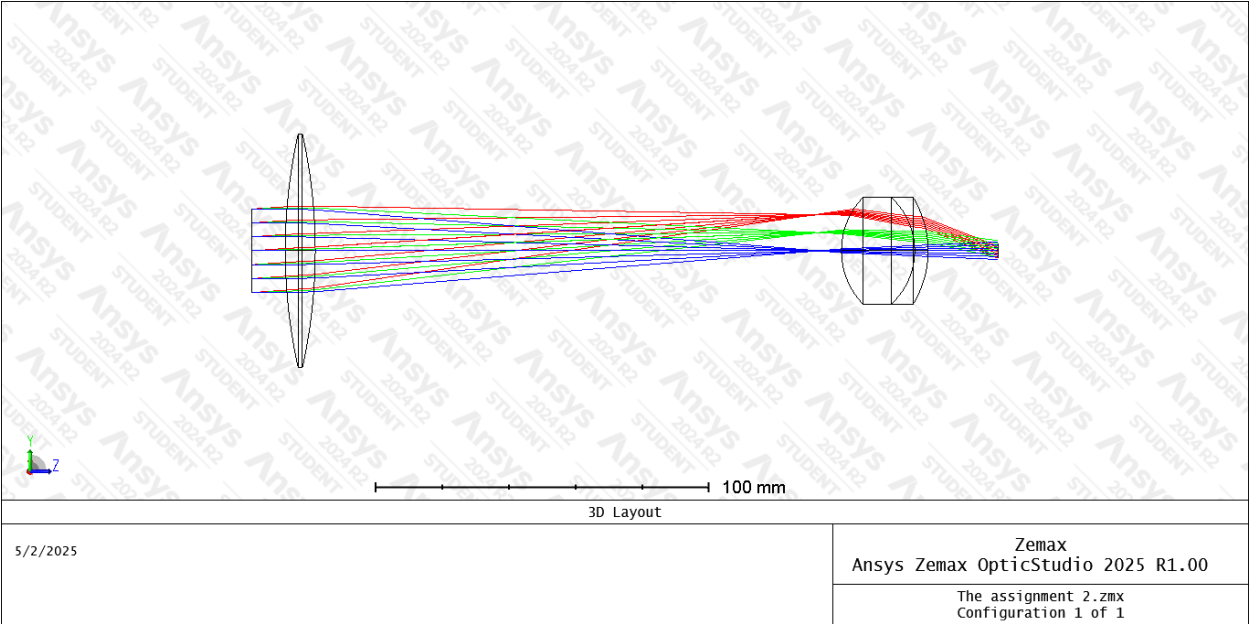
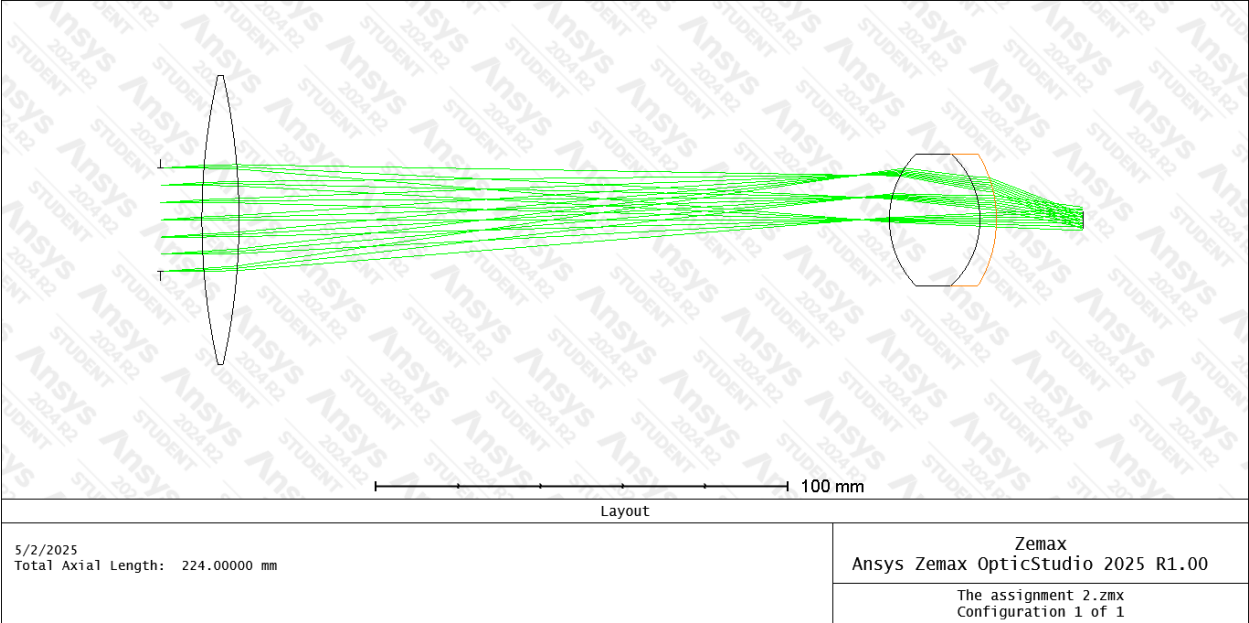
- Magnification: 6×
- Field of View (FOV): Analyze performance at 0°, 2°, and 4°
- Entrance Pupil Diameter: 25 mm

Aberration and Spot Analysis

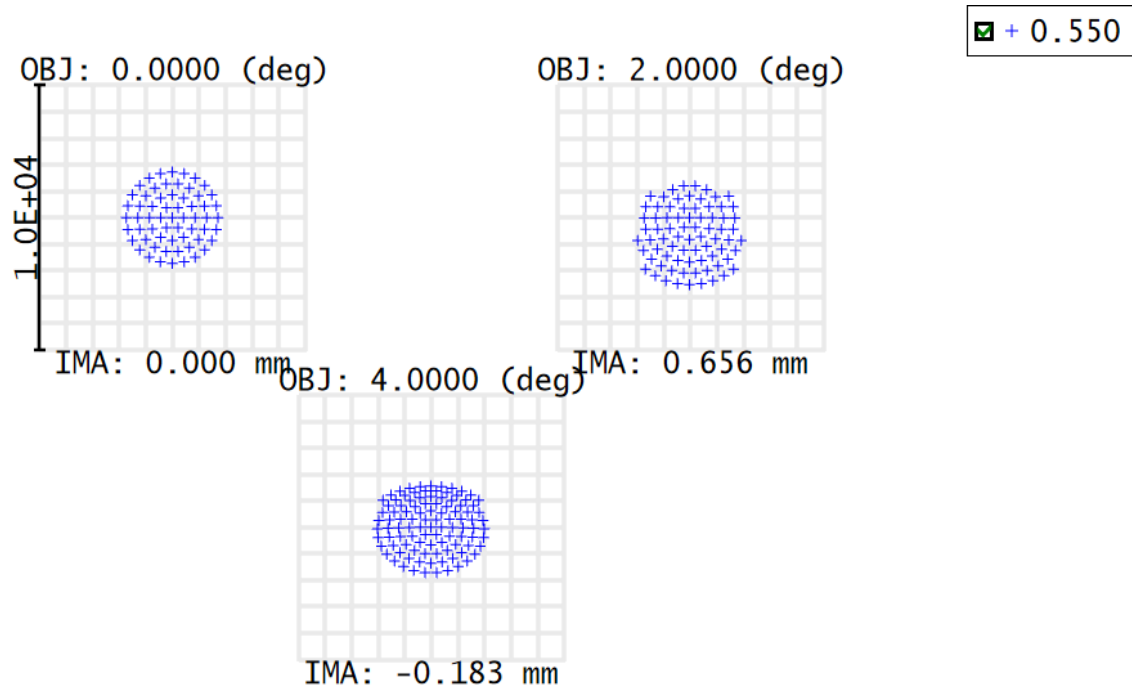
- Generate spot diagrams for 0°, 2°, and 4° fields
- Use Seidel coefficients or Ray Aberration plots to comment on dominant aberrations
- Note how FOV introduces off-axis aberrations (coma, astigmatism)

Wavefront Error Analysis

- Analyze Wavefront Map (OPD) at 0° and 4° field
- Compute RMS Wavefront Error
- Use Zernike polynomial coefficients to interpret wavefront shape



1) Spot Diagram



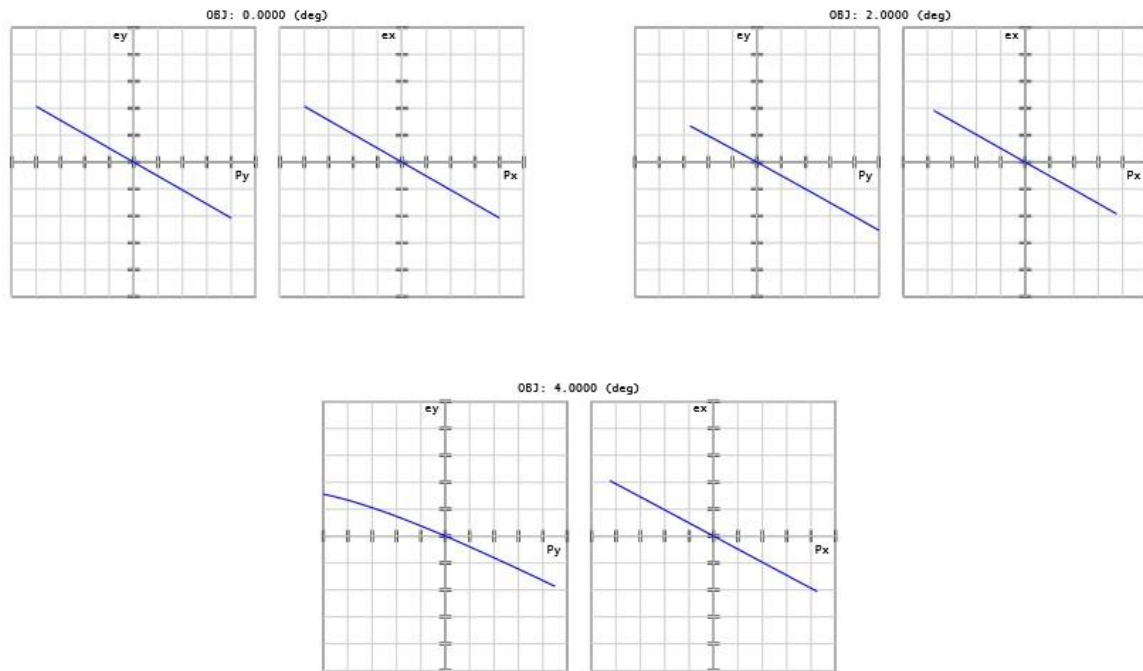
Surface: IMA

Spot Diagram		
5/2/2025		Zemax
Units are μm . Legend items refer to Wavelengths		Ansys Zemax OpticStudio 2025 R1.00
Field :	1 2 3	
RMS radius :	1354.06 1628.76 1443.21	
GEO radius :	1727.64 2554.17 2072.33	
Scale bar :	1.0E+04	Reference : Chief Ray
		assignment 7.zmx Configuration 1 of 1

The Spot diagram above for the fields 0 2 and 4 degrees with an average rms Radius of 1475.32 indicates the aberration distortion and coma and spherical aberration

2) Ray Fan

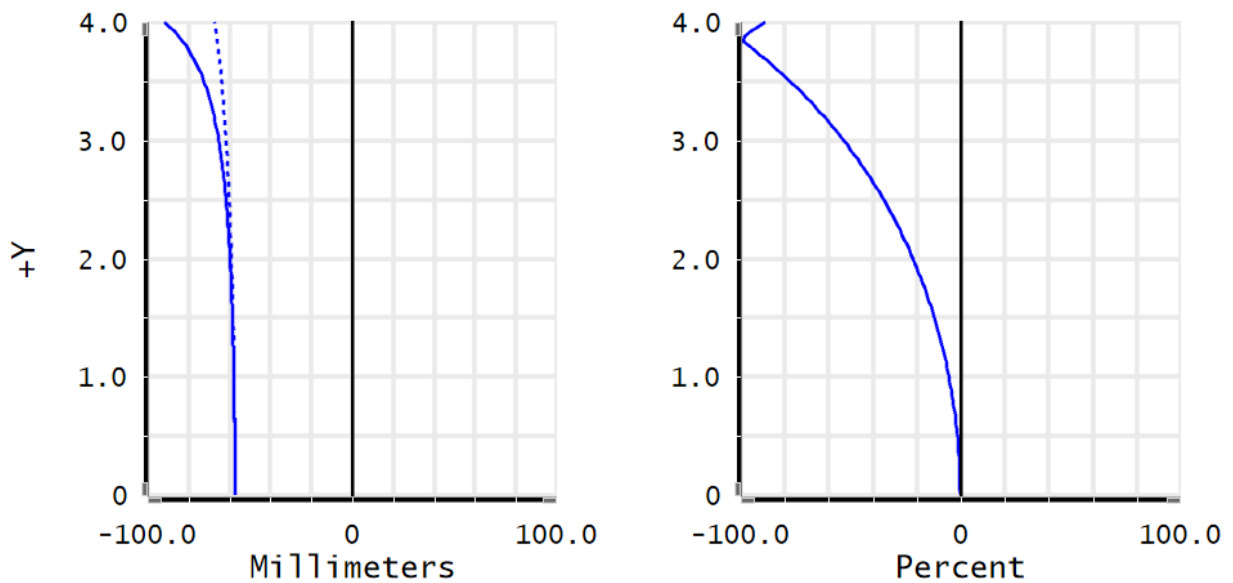
Ray fan for all of the fields are indicating the signs of defocus, coma and spherical aberration



Transverse Ray Fan Plot	
5/2/2025 Maximum Scale: $\pm 5000.000 \mu\text{m}$. <u>0.550</u> Surface: Image	Zemax Ansys Zemax OpticStudio 2025 R1.00 assignment 7.zmx Configuration 1 of 1

3) Field curvature and Distortion

The graph below shows the dominant aberration being distortion of about 99%



☒ 0.5500-Tangential ☒ 0.5500-Sagittal

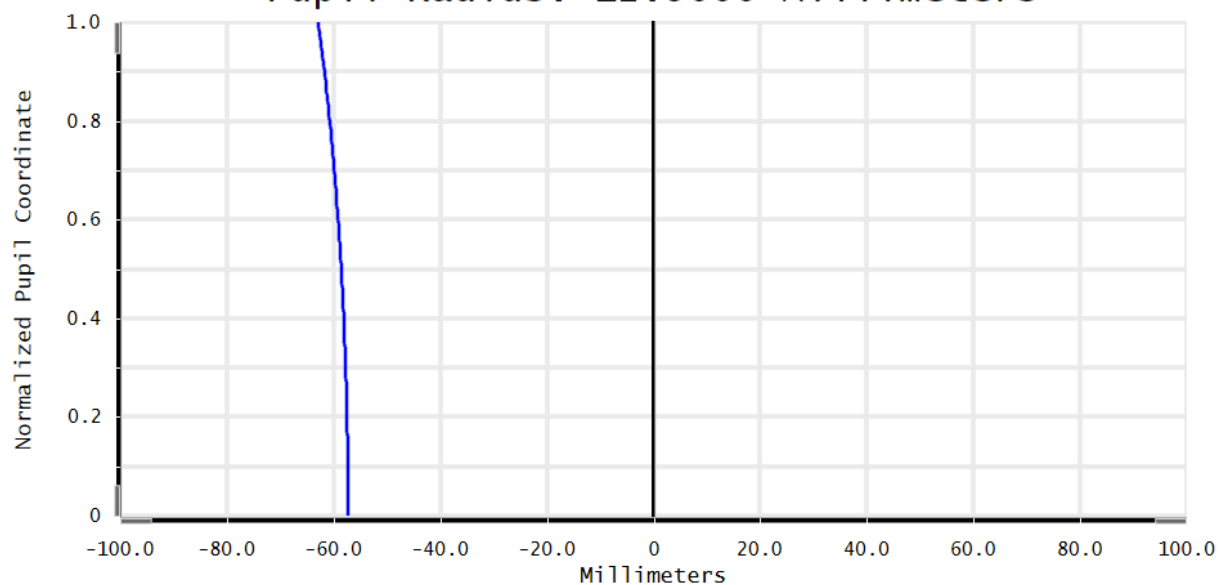
☒ 0.5500

Field Curvature	F-Tan(Theta) Distortion	
5/2/2025 Maximum Field is 4.000 Degrees. Field Curvature Sagittal = 9.8261 Millimeters Field Curvature Tangential = 34.7971 Millimeters Legend items refer to Wavelengths	5/2/2025 Maximum Field is 4.000 Degrees. Maximum distortion = 99.0555%	Zemax Ansys Zemax OpticStudio 2025 R1.00 assignment 7.zmx Configuration 1 of 1

4) Longitudinal Aberration

From the graph below it is clear that the rays are not symmetric around Y axis which is an indication of spherical aberration and also shows a significant longitudinal aberration

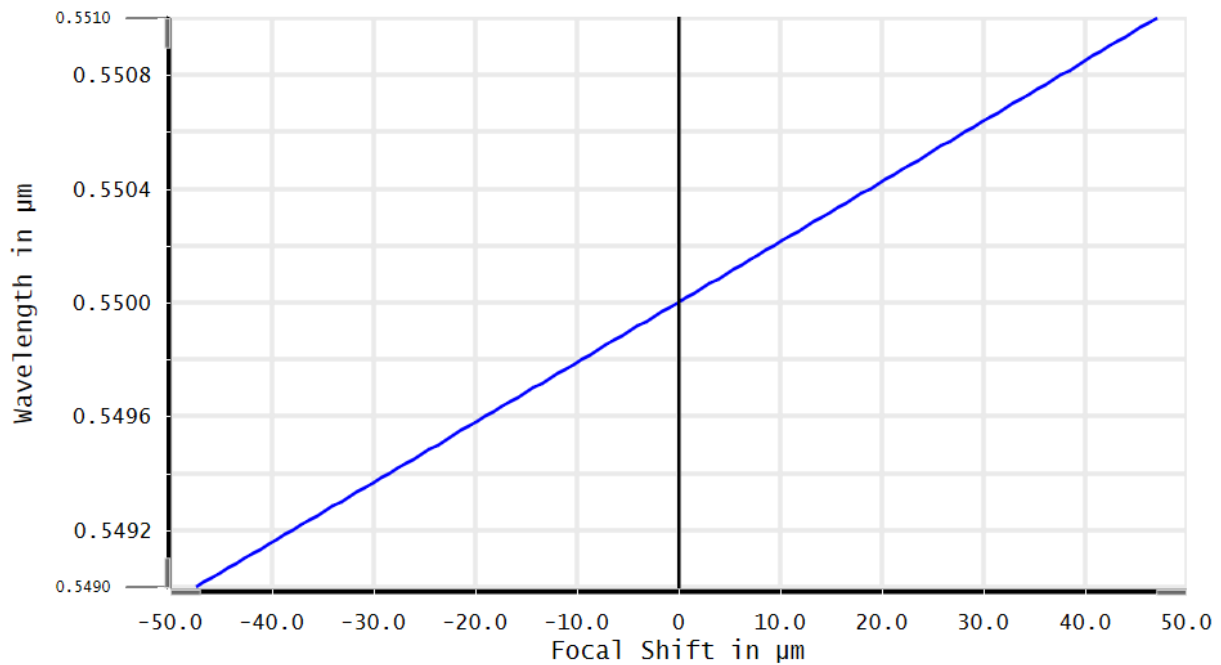
Pupil Radius: 12.5000 Millimeters



☒ 0.550

Longitudinal Aberration	
5/2/2025 Wavelengths: From 0.550 To 0.550 μm Legend items refer to Wavelengths	Zemax Ansys Zemax OpticStudio 2025 R1.00
	assignment 7.zmx Configuration 1 of 1

5) Chromatic Focal Shift

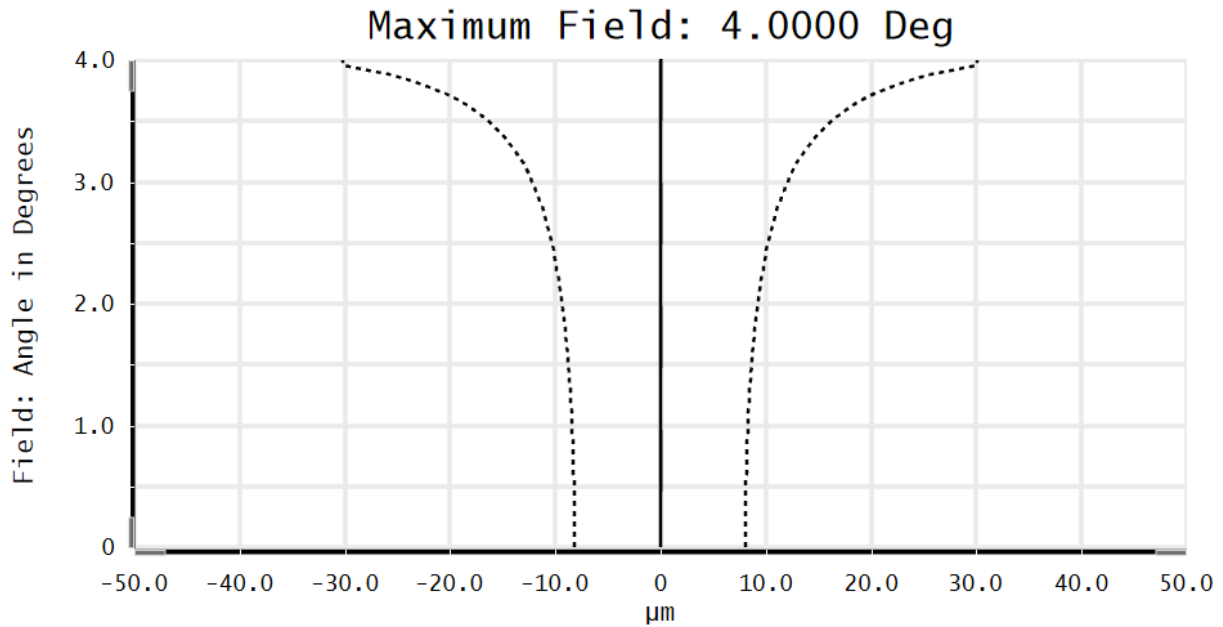


Chromatic Focal Shift	
5/2/2025 Maximum Focal Shift Range: 94.6320 μm Diffraction Limited Range: 322.110 μm Pupil Zone: 0.0000	Zemax Ansys Zemax OpticStudio 2025 R1.00
	assignment 7.zmx Configuration 1 of 1

A noticeable chromatic aberration can be seen above as the focus shifts linearly

5) Seidal Diagram

The Seidal Diagram shows the dominant aberration being distortion followed by spherical aberration and field curvature

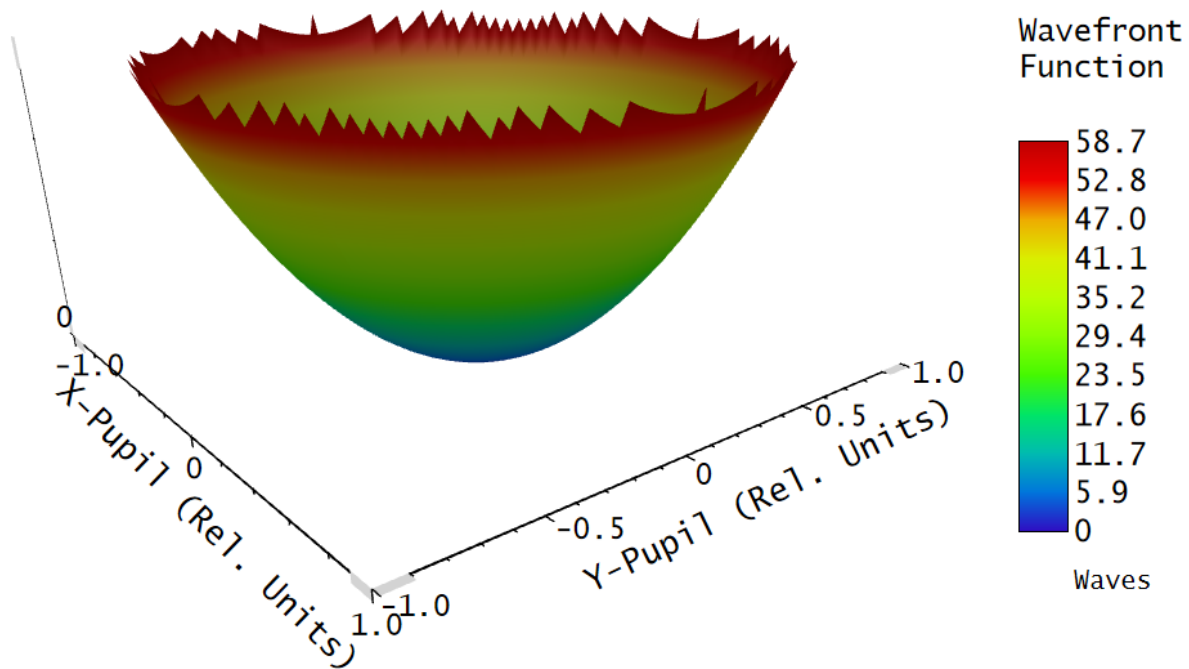


☒ — Shortest-Longest ☒ --- Airy

Lateral Color	
5/2/2025 Short Wavelength: 0.5500 μm Long Wavelength : 0.5500 μm Real rays used.	Zemax Ansys Zemax OpticStudio 2025 R1.00
	assignment 7.zmx Configuration 1 of 1

7) Wavefront Map Analysis

According to the wavefornt map the Zernike polynomial coefficient which is higher is th e Z4 which shows the aberration of astigmatism and Z1 which conveys coma spherical aberration and piston



Wavefront Function	
5/3/2025 0.5500 μm at 0.0000 (deg) Peak to valley = 58.7184 waves, RMS = 16.8546 waves. Surface: Image Exit Pupil Diameter: 4.3106E+00 Millimeters	Zemax Ansys Zemax OpticStudio 2025 R1.00
	LENS.zmx Configuration 1 of 1